



B.S. Abdur Rahman
Crescent
Institute of Science & Technology
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TOOL STUDY

OWASP ZAP : Open-Source Web Application Security Scanner for
Vulnerability Detection

Submitted By

KADHAMBARI DEVI S P

MAGESHWARI B

RRN : 230181601022

RRN : 230181601026

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**CSDX 409 PENETRATION TESTING AND
VULNERABILITY ASSESSMENT**

VI SEMESTER

Submitted To

Dr. T.Pandiyavathi

Assistant Professor (Sr. Gr.)

CSE DEPARTMENT

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INTRODUCTION

The purpose of this study is to understand and perform vulnerability assessment on a web application using OWASP ZAP. Web applications are widely used but are often exposed to various security threats such as hacking and data breaches. Identifying vulnerabilities is essential to improve application security.

OWASP ZAP (Zed Attack Proxy) is an open-source web application security scanner developed by the OWASP. It helps in detecting common vulnerabilities like SQL Injection, Cross-Site Scripting (XSS), and security misconfigurations. It acts as a proxy between the browser and the web application, allowing it to intercept and analyze network traffic.

TOOL OVERVIEW

OWASP ZAP is a widely used penetration testing tool designed for both beginners and professionals. It provides automated as well as manual testing features.

Features:

- Automated vulnerability scanning
- Spidering to discover web pages
- Active and passive scanning
- Alerts with severity levels
- Detailed reports generation
- User-friendly graphical interface for easy usage

Use Cases:

- Web application security testing
- Vulnerability assessment
- Ethical hacking and penetration testing
- Learning cybersecurity concepts

INSTALLATION AND SETUP

The OWASP ZAP tool was installed on a Windows system using the official installer.

System Details:

- Operating System: Windows
- Tool: OWASP ZAP
- Internet Connection: Required

Steps:

1. Downloaded OWASP ZAP from the official website
2. Installed using the Windows installer (.exe)
3. Launched the tool from the Start menu
4. Selected "Persist Session" option to begin

IMPLEMENTATION

The tool was used to perform an automated scan on a demo web application.

Target URL:

<https://demo.owasp-juice.shop>

Scan Type:

Automated Scan

Procedure:

1. Opened OWASP ZAP
2. Selected "Automated Scan"
3. Entered the target URL
4. Clicked on "Attack" to start scanning
5. The tool performed spidering and active scanning
6. Results were displayed in the Alerts section

RESULTS AND SCREENSHOT

The OWASP ZAP scan identified multiple vulnerabilities in the web application. The results were displayed in the Alerts section with severity levels.

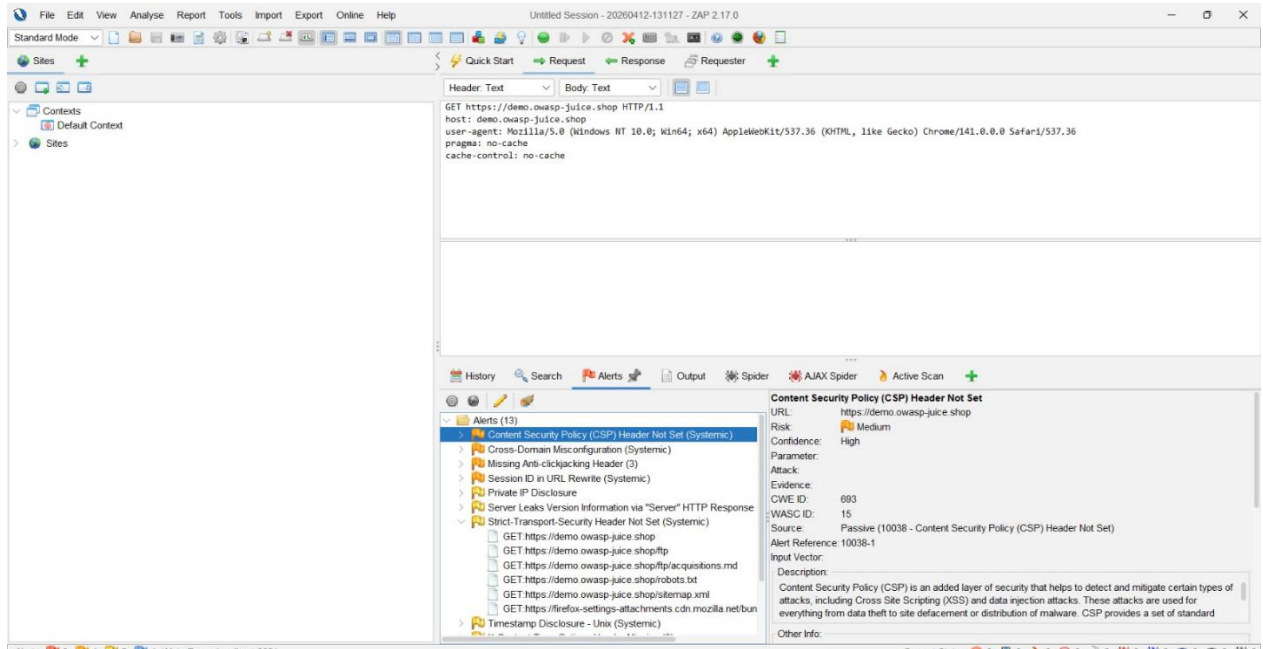


Figure 1: OWASP ZAP Scan Results

The above figure shows the vulnerabilities detected by OWASP ZAP. It displays various security issues along with their severity levels and descriptions.

VULNERABILITY FINDINGS

The following vulnerabilities were identified during the scan:

Vulnerability Name	Risk Level	Recommendation
CSP Header Not Set	Low	Configure Content Security Policy
Cross-Domain Misconfiguration	Medium	Restrict cross-origin access
Server Leaks Version Information	Low	Hide server details
Strict-Transport-Security Not Set	Low	Enforce HTTPS connections
X-Content-Type-Options Missing	Low	Set proper content-type headers

CONCLUSION

This study helped in understanding how OWASP ZAP can be effectively used to identify vulnerabilities in web applications. Through this tool, it was possible to analyze different types of security issues such as missing security headers and misconfigurations. The automated scanning feature made the process simple and efficient, even for beginners.

OWASP ZAP provides detailed information about each vulnerability, including its risk level, description, and possible solutions. This makes it easier to understand the impact of security flaws and take appropriate measures to fix them. The tool's user-friendly interface and clear output make it suitable for students and professionals who are new to cybersecurity.

Overall, OWASP ZAP is an effective tool for performing vulnerability assessment and improving the security of web applications. It not only helps in identifying weaknesses but also enhances knowledge about web security concepts and best practices.